

Electronic supplementary material

Large language model agents in a collective-risk social dilemma:
capability enables risk sensitivity but does not confer it

Every quantity below is regenerated from the raw per-game and per-probe logs by the scripts in `paper/revision/`; none is transcribed by hand.

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1 Vietnamese prompt: line-by-line back-translation

The Vietnamese decision template was prepared by a native speaker and holds the English structure line for line. Both files are thirty lines and carry the identical multiset of 42 substitution slots, and neither writes any game parameter as a literal, so the two languages cannot disagree about a value; this is checked mechanically by `r5_translation_qa.py`. The back-translations below are literal rather than idiomatic: they are meant to expose drift, not to read well. A dagger (†) marks a segment carrying a number or a conditional, the two things most at risk in translation.

English (original)	Vietnamese (as run)	Back-translation
† At the start, each player privately receives an endowment of <code>{endowment}</code> monetary units. Your final cash payoff depends on the rules below. The game lasts exactly <code>{nRounds}</code> rounds.	Khi bắt đầu, mỗi người chơi được nhận riêng một khoản vốn <code>{endowment}</code> đơn vị tiền. Số tiền mặt cuối cùng bạn thực nhận phụ thuộc vào các quy tắc dưới đây. Trò chơi kéo dài đúng <code>{nRounds}</code> vòng.	At the start, each player privately receives a capital of <code>{endowment}</code> money units. The final cash you actually receive depends on the rules below. The game lasts exactly <code>{nRounds}</code> rounds.

English (original)	Vietnamese (as run)	Back-translation
Each round, all <code>{nPlayers}</code> players decide simultaneously and in private how much to put into a shared “climate account”. The only amounts allowed are <code>{contributionOptions}</code> .	Mỗi vòng, tất cả <code>{nPlayers}</code> người chơi đồng thời và riêng tư quyết định đóng bao nhiêu vào một “quỹ khí hậu” chung. Các mức được phép chỉ gồm <code>{contributionOptions}</code> .	Each round, all <code>{nPlayers}</code> players simultaneously and privately decide how much to contribute to a shared “climate fund”. The permitted amounts consist only of <code>{contributionOptions}</code> .
† The group’s target: by the end of the game, the combined contributions of all <code>{nPlayers}</code> players must reach at least <code>{target}</code> (an average of <code>{fairShare}</code> per player per round).	Mục tiêu của nhóm: đến cuối trò chơi, tổng đóng góp của cả <code>{nPlayers}</code> người chơi phải đạt ít nhất <code>{target}</code> (trung bình <code>{fairShare}</code> mỗi người mỗi vòng).	The group’s target: by the end of the game, the total contribution of all <code>{nPlayers}</code> players must reach at least <code>{target}</code> (on average <code>{fairShare}</code> per person per round). <i>The equal-split hint is present in both languages and therefore cannot generate a language effect.</i>
† If the group reaches <code>{target}</code> : every player keeps, in cash, whatever they did not contribute (for example, contributing <code>{fairShare}</code> every round leaves you <code>{fairShareKept}</code> at the end).	Nếu nhóm đạt <code>{target}</code> : mỗi người giữ lại (bằng tiền mặt) toàn bộ số tiền mình không đóng góp (ví dụ, đóng <code>{fairShare}</code> mỗi vòng thì cuối cùng bạn còn <code>{fairShareKept}</code>).	If the group reaches <code>{target}</code> : each person keeps (in cash) all the money they did not contribute (for example, contributing <code>{fairShare}</code> every round leaves you <code>{fairShareKept}</code> at the end).
† If the group does not reach <code>{target}</code> : the computer runs a lottery: with probability <code>{riskPercent}%</code> the group is hit by a disaster and every player loses all of their remaining money (final cash payoff 0); with the remaining <code>{safePercent}%</code> probability, each player keeps their remaining money.	Nếu nhóm không đạt <code>{target}</code> : máy tính quay xổ số: với xác suất <code>{riskPercent}%</code> nhóm gặp thảm họa và mọi người chơi mất tất cả số tiền còn lại (tiền mặt cuối cùng = 0); với xác suất <code>{safePercent}%</code> còn lại, mỗi người giữ được số tiền còn lại của mình.	If the group does not reach <code>{target}</code> : the computer draws a lottery: with probability <code>{riskPercent}%</code> the group meets a disaster and every player loses all remaining money (final cash = 0); with the remaining <code>{safePercent}%</code> probability, each person keeps their remaining money. <i>The critical conditional: antecedent, both branches, both percentages and the zero payoff are preserved in the same order.</i>
Money put into the climate account is gone for good and is never refunded, whether or not the target is reached.	Tiền đã bỏ vào quỹ khí hậu là mất hẳn, không bao giờ được hoàn lại, dù có đạt mục tiêu hay không.	Money put into the climate fund is lost for good, never refunded, whether or not the target is reached.

English (original)	Vietnamese (as run)	Back-translation
After each round, all <code>{nPlayers}</code> contributions for that round are revealed to everyone. Names are hidden, but each player always appears in the same fixed position. Players cannot talk to or message each other.	Sau mỗi vòng, đóng góp của cả <code>{nPlayers}</code> người trong vòng đó được hiển thị cho mọi người. Danh tính được ẩn, nhưng mỗi người luôn xuất hiện ở cùng một vị trí cố định. Người chơi không được trao đổi hay nhắn tin cho nhau.	After each round, the contributions of all <code>{nPlayers}</code> people in that round are displayed to everyone. Identities are hidden, but each person always appears in the same fixed position. Players may not converse or message each other.
† Current state: Round <code>{currentRound}</code> of <code>{nRounds}</code> . Your remaining money: <code>{remainingEndowment}</code> .	Tình trạng hiện tại: Vòng <code>{currentRound}</code> trên tổng <code>{nRounds}</code> . Số tiền còn lại của bạn: <code>{remainingEndowment}</code> .	Current situation: Round <code>{currentRound}</code> of a total <code>{nRounds}</code> . Your remaining money: <code>{remainingEndowment}</code> .
† The climate account so far holds <code>{groupAccount}</code> of the <code>{target}</code> target. (<i>printed-total arm only</i>)	Quỹ khí hậu đến lúc này có <code>{groupAccount}</code> trên mục tiêu <code>{target}</code> .	The climate fund so far has <code>{groupAccount}</code> of the <code>{target}</code> target.
† Now decide your contribution for round <code>{currentRound}</code> . Allowed choices: <code>{contributionOptions}</code> . Output only your decision as a final line: CONTRIBUTION: <one of <code>{contributionOptions}</code> >	Bây giờ hãy quyết định mức đóng góp của bạn cho vòng <code>{currentRound}</code> . Các lựa chọn hợp lệ: <code>{contributionOptions}</code> . Chỉ xuất ra quyết định của bạn ở dòng cuối: CONTRIBUTION: <một trong các giá trị <code>{contributionOptions}</code> >	Now decide your contribution for round <code>{currentRound}</code> . Valid choices: <code>{contributionOptions}</code> . Output only your decision on the final line. <i>The answer token CONTRIBUTION: is left untranslated in both, so parsing is language-independent.</i>

The round history is built by the engine rather than the template, and is localised alongside it: (you) $\rightarrow$ (bạn), Round $n \rightarrow$ Vòng n , you contributed c ; group round total $t \rightarrow$ bạn đóng c ; tổng nhóm vòng này t , and (cumulative c) $\rightarrow$ (tích lũy c). No English leaks into a Vietnamese prompt.

1.1 Behavioural validation of the translation

If the Vietnamese prompt had corrupted a number, a rule or the conditional, the agents reading it would answer questions about that content worse. Table 2 shows they do not. The gap is confined to the *State* axis, which asks for quantities the agent must compute rather than read, and to one model on one question.

The one Rules question with a gap is a property of a single model, not of the translation. On `rules_target`, Qwen2.5-7B, Gemma-2-9B, Gemma-2-27B, Qwen2.5-32B, Llama-3.3-70B and Qwen2.5-72B all answer at 100% in both languages; Llama-3.1-8B falls from 100% to 37.2%. A corrupted prompt would degrade every reader of it. Llama-3.1-8B is also the only model with a non-negligible probe parse-failure rate, 2.05% of its Vietnamese probes.

Tokenisation. Measured on the rendered prompts logged in `turns.jsonl`, matched by round, with `o200k_base` as a proxy for the hosted tokenisers, the mean prompt is 537.4 tokens in English

Table 2: Comprehension accuracy by language. *Rules* covers the endowment, target, action set, round count, catastrophe probability and the payoff under catastrophe.

Probe	English	Vietnamese	Gap (pp)
Rules axis, all questions (20,160 probes)	99.40%	98.74%	+0.66
rules_risk_pct (state the catastrophe probability)	98.02%	98.97%	−0.95
rules_payoff_disaster	100.00%	100.00%	0.00
rules_actions, rules_endowment, rules_n_rounds	100.00%	100.00%	0.00
rules_min_contrib	100.00%	99.92%	+0.08
rules_max_contrib	97.22%	100.00%	−2.78
rules_target	100.00%	91.03%	+8.97
Time axis (197,400 probes)	95.65%	93.17%	+2.48
State axis (163,800 probes)	73.63%	61.80%	+11.83

and 642.6 in Vietnamese, a ratio of 1.203. Both are far inside the 4,096-token context used for the open-weight arm, so nothing is truncated in either language.

2 Decoding parameters and run-to-run stability

Table 3: What was sent to the hosted models. Read from the task source (`kaggle/benchmarks/crg_task_server.py`) rather than transcribed.

Parameter	Value	Set by
temperature	0.7, every request	us (matches the open-weight arm)
per-turn seed	$(\text{base} + \text{rep}) \cdot 10^5 + \text{round} \cdot 100 + \text{agent}$	us (honouring not guaranteed)
top_p	not set	provider default, undocumented
max_completion_tokens	512 non-reasoning; 6,000 reasoning	us
Grok-4.20 (no reasoning)	6,000 and 8,000 across shards	us
Claude-Opus-5	6,000 and 16,000 across shards	us
parse retries	3, with a fresh seed each attempt	us

Accidental replication. Six games were executed twice, hours apart in a separate session and in four cases on a different Kaggle account, because the smoke pass and a path-length download failure both left duplicate coverage. Five reproduced the group total exactly; one differed by 4 points of 240 (Claude-Opus-5, $p = 0.1$, Vietnamese: 124 versus 120).

Configuration	Cell	Group totals	Sources
Claude-Opus-5	$p=0.1$, vn	124, 120	two accounts
Claude-Opus-5	$p=0.9$, en	128, 128	smoke pass, production shard
Gemini-3.1-Pro	$p=0.9$, en	120, 120	smoke pass, production shard
GPT-5.6-sol	$p=0.9$, en	120, 120	smoke pass, production shard
Grok-4.20 (no reasoning)	$p=0.1$, vn	240, 240	two accounts
Grok-4.20 (reasoning)	$p=0.9$, en	120, 120	smoke pass, production shard

Execution-batch test. Every Claude-Opus-5 cell was assembled from two to four separate executions. A one-way permutation ANOVA (10,000 permutations) of group contribution on

execution batch, within cell, finds no effect in any of the twelve cells that span more than one batch: $F \leq 1.91$, all $P > 0.22$. Eleven of those splits were accidents of sharding; the twelfth, the $p = 0.3$ cell added later (§6), was split deliberately, five games on each of two accounts, and gives $F = 0.40$, $P = 1.00$.

3 The lottery cannot induce learning

The catastrophe is drawn once, after round ten (`crsd/engine/scoring.py::compute_outcome`), and its outcome is never shown to any agent; agents carry no state between games. The following are therefore confirmations, not tests of a live hypothesis.

Table 4: No drift across repetitions, and no response to a previous repetition’s catastrophe. Contributions are centred within (model, risk, language) cell so the coefficients are pure within-cell contrasts. Units are points of 240.

Test	Arm	Coefficient	SE	P
Contribution on repetition index	open-weight	+0.033	0.103	0.747
	commercial	−0.030	0.195	0.876
Contribution on previous repetition’s catastrophe	open-weight	+0.022	1.367	0.987
	commercial	−2.500	2.527	0.323

Independent per-game lottery. Because the lottery never touches behaviour, the alternative implementation is a recomputation over the observed contribution vectors. Redrawing an independent uniform per game, 10,000 times, at $p = 0.1$ in English: Gemini-3.1-Pro earns 36.0 per agent (95% interval [28.0, 40.0]), GPT-5.6-sol 35.7 ([27.8, 39.7]). Every configuration that reached the target faces no lottery at all and earns its deterministic amount, 20.0 for Claude-Opus-5 and for Grok-4.20 with reasoning.

4 Within-game adaptation

Let $\text{need}(r) = (120 - S_r)/(6(11 - r))$ be the per-agent contribution still needed at the start of round r , where S_r is the pool before that round. We regress an agent’s contribution on need, its own mean contribution in earlier rounds (which removes the mechanical path by which an agent’s own generosity lowers need), and round fixed effects. The composition study is the right arm for this: the standard deviation of need is 2.18 there against 1.66 in the baseline, where almost every group is on pace at almost every round.

Table 5: Pace tracking in the composition study. A coefficient of 1.0 on need would be an agent that exactly covers the remaining shortfall; 0 an agent that ignores it. The interaction is the reviewer’s question: does falling behind matter more when the catastrophe is more likely?

Model	need	P	need $\times$ risk	P
Gemma-2-9B	−0.070	$< 10^{-4}$	−0.005	0.311
Llama-3.1-8B	−0.083	$< 10^{-4}$	+0.018	0.531
GPT-5.4-nano	−0.093	$< 10^{-4}$	−0.018	0.160
Qwen2.5-7B	+0.438	$< 10^{-4}$	−0.022	0.480

Endgame push. Of the composition games, 548 needed more than 2 per agent per round at the start of round eight. Comparing each group’s mean contribution in rounds nine and ten with its own mean in rounds five to seven, the push is −0.151 at $p = 0.1$ ($n = 183$), −0.108 at $p = 0.5$

($n = 181$) and -0.056 at $p = 0.9$ ($n = 184$). Regressing the push on risk gives $+0.119$ per unit probability (SE 0.061, $P = 0.051$): about 1.1 points of 240 across the full risk sweep.

5 How strong is the equal-split anchor?

Table 6: Probability of taking the focal action named by the prompt (contribute 2), over all baseline turns, against the size of the model’s English risk effect. If anchoring produced the risk null, models that ignore the anchor would be free to respond to risk. The correlation between the two columns is -0.06 .

Configuration	$P(\text{contribute 2})$	$P(0)$	$P(4)$	$ \Delta \text{ risk} $
Grok-4.20 (reasoning)	1.000	0.000	0.000	0.2
Gemini-3.1-Flash-Lite	0.991	0.001	0.008	1.8
Gemma-2-27B	0.946	0.001	0.053	3.8
Gemma-2-9B	0.926	0.000	0.074	0.0
Qwen2.5-72B	0.896	0.000	0.104	2.4
Qwen2.5-32B	0.880	0.028	0.092	1.4
Claude-Opus-5	0.868	0.047	0.085	8.0
Llama-3.1-70B	0.866	0.063	0.071	0.4
GPT-5.6-sol	0.706	0.281	0.013	118.2
Gemini-3.1-Pro	0.667	0.333	0.000	120.0
Llama-3.1-8B	0.282	0.086	0.631	3.4
GPT-5.4-nano	0.246	0.309	0.445	7.2
Grok-4.20 (no reasoning)	0.118	0.003	0.879	23.6
Qwen2.5-7B	0.054	0.003	0.943	0.0

How far each manipulation moves $P(\text{contribute 2})$, as the median range across the models run in that arm: catastrophe risk 3.58 percentage points in the baseline and 1.47 in the composition study; language 9.80; composition from zero to six selfish agents 14.67. At $p = 0.1$ in English, Gemini-3.1-Pro contributes 0 in 100.0% of its turns and GPT-5.6-sol in 98.8%, overriding the focal action entirely.

6 The five-point risk grid, cell by cell

The four top-tier configurations were re-run in English at $p = 0.3$ and $p = 0.7$, ten games per cell, to locate the switch that the original three levels could only bracket. Table 7 gives every cell of the resulting grid. The expected-value solution contributes 0 at $p = 0.1$ and $p = 0.3$, is indifferent at $p = 0.5$, and contributes 120 at $p = 0.7$ and $p = 0.9$.

The added cells were executed on separate Kaggle accounts from the original sweep, and the $p = 0.3$ Claude-Opus-5 cell was deliberately split five games to one account and five to another as a further check on that source of variation; a permutation ANOVA on execution batch within that cell gives $F = 0.40$, $P = 1.00$. Contributions parsed without failure in all 80 added games, that is, in all 4,800 added agent-turns.

Table 7: Group contribution of a possible 240 (standard deviation across the ten games in parentheses) and target-reach rate, in English, at five catastrophe risks. Only GPT-5.6-sol at $p = 0.3$ is an intermediate cell: seven of its ten groups reached the target and three did not, with pooled totals of 114, 120 and 122. Its switch therefore falls at or below $p = 0.3$, one grid step earlier than risk neutrality requires, whereas Gemini-3.1-Pro’s falls between $p = 0.3$ and $p = 0.5$, where risk neutrality puts it.

Configuration		$p=0.1$	$p=0.3$	$p=0.5$	$p=0.7$	$p=0.9$
Gemini-3.1-Pro	contribution	0.0 (0)	4.2 (3.8)	120.0 (0)	120.0 (0)	120.0 (0)
	reach	0%	0%	100%	100%	100%
GPT-5.6-sol	contribution	1.6 (1.5)	118.4 (2.9)	120.0 (0)	120.0 (0)	119.8 (0.6)
	reach	0%	70%	100%	100%	90%
Claude-Opus-5	contribution	120.0 (0)	120.6 (0.9)	126.8 (2.7)	128.0 (1.8)	128.0 (2.4)
	reach	100%	100%	100%	100%	100%
Grok-4.20 (reasoning)	contribution	120.0 (0)	120.0 (0)	120.0 (0)	120.0 (0)	120.2 (0.6)
	reach	100%	100%	100%	100%	100%